GatedVLA with Camera-Based Emergency Braking: AlpaSim E2E Closed Loop Challenge Technical Report

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Abstract

This report presents GatedVLA, our entry for the Physical AI AV (PAI-AV) track. Our learned vision-language-action (VLA) policy pairs the driving-pretrained Qwen-Drive-1.0-4B vision-language model [20] with a layer-wise flow-matching action expert that outputs a 2.5 s horizon of acceleration and curvature controls, augmented with a soft visual mask applied to the expert's cross-attention. Compared to an identically trained baseline, the masked policy achieves comparable overall open-loop error while reducing error on left turns. In local closed-loop evaluation across the same 150 scenes (two rollouts per scene), the visual mask increases the mean scene score from 0.479 to 0.517 (paired difference $+0.038$, 95% bootstrap CI $[-0.018, +0.093]$, wins/losses 37/26, sign test $p = 0.21$). Incorporating a rule-based automatic emergency braking (AEB) module at deployment further increases the mean score to 0.599 ($+0.082$, CI $[+0.023, +0.143]$) and reduces the mean number of scenes with at-fault collisions from 34.0 to 11.0.

1 Introduction

Even minor planning errors alter the subsequent observations encountered by a driving policy—a compounding effect that only closed-loop evaluation can reveal. The AlpaSim challenge evaluates policies in closed loop on held-out scenes [12], where a scene receives a score of zero if the ego vehicle causes an at-fault collision, drives off-road, or exits the recorded driving corridor. In our preliminary experiments, at-fault collisions were the primary source of score loss, predominantly involving slower or stationary lead vehicles when the planned trajectory failed to initiate braking. We therefore introduce (1) a learned visual mask trained to selectively retain image evidence relevant to navigation intent for the action expert, and (2) a detector-based emergency braking module that intervenes when the learned policy fails to decelerate.

2 Related work

Vision-language-action driving. Qwen-Drive-1.0 [20] establishes a vision-language foundation model for autonomous driving paired with a planning expert. Alpamayo-R1 [13] couples a reasoning model with a diffusion-based trajectory decoder, while DriveMoE [17] introduces expert routing to Drive-π_0, an adaptation of π_0 [2]. We train a π_0-style flow-matching expert on top of the Qwen-Drive backbone. **Limiting visual evidence.** S2 [16] learns per-token soft masks under a visual evidence budget alongside an ungated pathway. We adopt a similar gating head and budget formulation, but apply the mask directly to the action expert's cross-attention logits, enforce an exact budget by construction, and train the mask via self-distillation (Section 4). **Copycat behavior.** Imitation learning policies can suffer from causal confusion by over-relying on their own past motion [4]; ChauffeurNet [1] mitigates this with history dropout during training. We explored both history dropout and temporal ensembling [18], but observed no clear gains (Appendix H).

3 Method summary

Our driving system (Fig. 1) comprises an end-to-end learned policy that replans every 0.5 s and a detector-based AEB module that exclusively decelerates the planned trajectory along its predicted path. We evaluated and submitted three deployment variants derived from a single trained checkpoint: AEB at 10 Hz, AEB at 5 Hz, and the policy without AEB. Our implementation builds upon starVLA [15] and py123d_garage [3, 5], and all evaluations are conducted in AlpaSim [12].

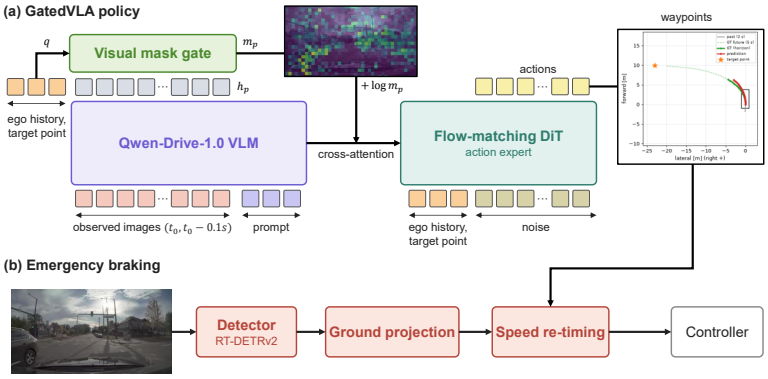


Figure 1: System overview. (a) GatedVLA policy, trained end-to-end: the visual mask m_p biases the expert's cross-attention to the VLM's image tokens. (b) Pretrained detector and rule-based AEB, integrated only at deployment: they re-time the planned waypoints along their path.

4 Policy description

Inputs. Two front wide-angle camera frames (at t_0 and $t_0 - 0.1$ s) along with the fixed text prompt "predict trajectory" are fed into the VLM. Ego-state history and navigation conditioning are provided exclusively to the action expert: five 10 Hz historical steps of $[x, y, \text{heading}, \text{speed}, a_{\text{lon}}]$ concatenated at each step with a target point $(x_{\text{tp}}, y_{\text{tp}})$ and a binary target-presence flag. During training, the target point is the first future ego position at a Euclidean distance of at least 25 m; at deployment, it is the first such point along AlpaSim's map route (Appendix E).

Expert and outputs. The action expert follows a π_0-style layer-wise architecture [2, 17]: a 32-block Diffusion Transformer (DiT) [14] alternates self-attention and cross-attention layers, where each cross-attention block attends to a corresponding VLM layer via a learned linear projection. The expert is trained via conditional flow matching [8] with a velocity MSE objective and generates trajectories using 10 Euler integration steps at inference. It predicts 25 steps of $[a_{\text{lon}}, \text{curvature}]$ controls at 10 Hz, utilizing the kinematic unicycle action space adapted from NVIDIA Alpamayo 1.5 [10, 13]. Integrating these controls from the current ego speed yields the output poses.

Baseline. The baseline shares the same architecture and training protocol but omits the visual mask and Paths B and C (below); as in Path A, the target point is dropped with probability $p = 0.5$ per sample.

Visual mask. A gating module takes each visual token's detached final-layer VLM hidden state h_p alongside a conditioning query q (which embeds the scaled target point and ego state) and computes a logit score from $[h_p, q, h_p \odot q]$, following the visual evidence budget formulation of S2 [16]. The scores are standardized across tokens within each frame to obtain z_p, yielding soft mask values $m_p = \sigma(b + 2z_p)$, where the scalar bias b is solved via bisection such that the per-frame mean of m_p equals the target budget $\rho = 0.2$ by construction, eliminating the need for an auxiliary budget loss. We add $\log m_p$ directly to the expert's cross-attention logits while leaving text tokens and internal VLM representations unmasked. Each training step executes three expert passes sharing a single VLM forward pass, sampled noise, and flow time step:

- **Path A** (ungated, target point dropped with $p = 0.5$): minimizes the standard flow-matching loss and updates both the VLM and DiT.
- **Path B** (stop-gradient mask, target point provided): minimizes $\|v_B - \text{sg}(v_A)\|^2$ to update the DiT; this serves as the inference pathway.

- **Path C** (active mask, target point omitted from the DiT state but provided to the gate): minimizes $\|v_C - \mathrm{sg}(v_A)\|^2$ and updates only the gating module.

Paths B and C are distilled only on samples where the target point is retained in Path A; consequently, Path C forces the gating module to preserve visual tokens that encode the navigation intent.

Emergency brake. An RT-DETRv2-R18 detector [9, 19] pretrained on COCO [7] processes the current front-camera frame and retains boxes of seven traffic classes with confidence scores exceeding 0.5. The central 40% of each bounding box's bottom edge is projected onto the ego rig's flat ground plane using the nominal f-θ camera calibration. Among detected obstacles within a 1.3 m lateral corridor of the planned path, the closest obstacle defines the arc-length gap d measured from the front bumper along the path; the closing speed c is estimated via linear regression over the last 1 s of d (Table 4). With braking margin g and required deceleration a_{req},

$$g = d - 2.5\,\mathrm{m} - 0.4\,\mathrm{s} \cdot c, \qquad a_{\mathrm{req}} = \frac{c^2}{2\max(g,\, 0.1\,\mathrm{m})}, \tag{1}$$

the planned speed profile is capped by a linear deceleration ramp from the current speed v down to the estimated lead speed $\max(v - c, 0)$ at a rate of $\mathrm{clip}(1.2\,a_{\mathrm{req}}, 3, 8)\,\mathrm{m/s^2}$ whenever $g \le 0$ or $a_{\mathrm{req}} > 3\,\mathrm{m/s^2}$. Under a milder time-to-collision condition ($d/c < 3\,\mathrm{s}$), the planned speed is constrained not to exceed the current ego speed. Because the trajectory is re-timed along its own path from the ego vehicle's projection, a braked ego will not accelerate to catch up with a stale plan. Upon AEB release, the speed is held for one control tick before an immediate policy replan.

5 Resources and training

Data. Models are trained on the NVIDIA PhysicalAI-Autonomous-Vehicles dataset [11] (NVIDIA Autonomous Vehicle Dataset License Agreement), using only the front wide-angle camera streams and egomotion trajectories from the training split. No LiDAR, radar, or machine-generated perception labels were used.

Pretrained weights. We initialize the VLM from Qwen-Drive-1.0-4B [20] (Apache-2.0) and use off-the-shelf RT-DETRv2-R18 weights (`PekingU/rtdetr_v2_r18vd`, Apache-2.0) for detection. **Private resources:** None.

Training. Both the baseline (W&B run `ccb412c9`) and masked policy (`03bdfc50`) were trained from the same code commit with random seed 42 for 12 epochs (38,400 steps) across 12 Intel Aurora XPU tiles. In the spirit of linear probing then fine-tuning (LP-FT) [6], the VLM backbone remained frozen during the first epoch. Full hyperparameter settings and training curves are provided in the appendix.

Open-loop results. Table 1 reports open-loop trajectory prediction performance on a fixed validation subset of 240 samples (80 left turns, 80 right turns, and 80 straight segments). Overall open-loop error is comparable between the two models: the visual mask improves performance on left turns (FDE of 1.100 vs. 1.213; miss rate of 0.150 vs. 0.200) and yields a slightly lower FDE on right turns, at the cost of higher error on straight segments. Ablating visual inputs degrades FDE substantially more than omitting the navigation target point (masked policy: 1.564 with black frames vs. 1.135 with no target and no mask), confirming that visual evidence strongly informs planning in both models (baseline with black frames: FDE 1.408). Appendix D gives all controls.

Table 1: Open-loop prediction error at step 38,400 (baseline / mask). ADE and FDE in meters over 2.5 s; miss: any waypoint error > 2 m.

	ADE	FDE	Miss rate
All	0.337 / 0.341	1.029 / 1.023	0.125 / 0.125
Left	0.405 / 0.376	1.213 / 1.100	0.200 / 0.150
Right	0.367 / 0.360	1.136 / 1.089	0.087 / 0.100
Straight	0.239 / 0.287	0.737 / 0.879	0.087 / 0.125

Closed-loop results. Table 2 summarizes local closed-loop evaluation across the same 150 scenes sampled from the 441-scene curated validation split using the `dev` challenge preset (40 m route offset). A rollout without such a failure scores $\min(\mathrm{progress}/0.8, 1)$. The first and second rollouts ran on different hosts (NVIDIA Blackwell and Ada), so differences between them reflect both hardware and run-to-run variance. Averaging scores across the two rollouts per scene: mask vs. baseline

achieves a mean difference of $+0.038$ (95% bootstrap CI $[-0.018, +0.093]$, wins/losses 37/26, sign test $p = 0.21$); AEB at 10 Hz vs. mask alone yields $+0.082$ (CI $[+0.023, +0.143]$, 41/28, $p = 0.15$); and AEB at 5 Hz vs. 10 Hz yields -0.009 (CI $[-0.065, +0.044]$, 34/34, $p = 1.00$). Appendix G gives all pairwise comparisons.

Table 2: Local closed-loop results across the same 150 scenes. Failure counts are numbers of affected scenes; rear-end collisions by other agents are not at-fault and merely halt progress accrual. Mean Drive latency includes driver video recording.

Configuration	Host	Score	At-fault	Offroad	Corridor exit	Rear-ended	Drive (s)
Baseline, 1st	Blackwell	0.476	42	25	8	12	0.133
Baseline, 2nd	Ada	0.481	41	19	15	9	0.066
Mask, 1st	Blackwell	0.499	35	22	12	15	0.110
Mask, 2nd	Ada	0.534	33	23	8	15	0.146
Mask + AEB 10 Hz, 1st	Blackwell	0.607	12	27	11	25	0.201
Mask + AEB 10 Hz, 2nd	Ada	0.591	10	31	13	23	0.085
Mask + AEB 5 Hz	Ada	0.590	13	27	10	28	0.264

6 Runtime and reproducibility

Serving. The submission container image is self-contained, operates within the challenge sandbox without external network access, and launches without arguments. Between 0.5 s replans, it serves poses from the cached plan at 10 Hz.

Latency and memory. The mean `Drive` latencies reported in Table 2 were recorded locally with driver video recording enabled and up to 12 concurrent rollouts sharing a single driver on multi-tenant hosts, so they are only indicative. In a standalone run with a single rollout per replica and video recording disabled (on a shared NVIDIA RTX 6000 Ada GPU), the mean `Drive` RPC latency was 0.10–0.11 s for both AEB variants (Appendix I). The RT-DETRv2 detector requires approximately 0.2 GiB of VRAM in addition to the policy's 11.6 GiB footprint.

Reproducibility. The deployed inference configuration matches our logged W&B training runs. Source code: `https://github.com/Azuma413/GatedVLA` (branch `main`, commit `80ab247`).

7 Limitations and safety considerations

Limited generalization evidence for AEB. The AEB decision thresholds were tuned offline by replaying cached detections on the same 150 scenes evaluated in Table 2, so the observed gains may be optimistic for held-out private test scenes.

Remaining collisions. Of the 12 remaining at-fault collisions in the first 10 Hz AEB rollout (manual inspection, Appendix G), 6 involved obstacles offset by approximately 2 m laterally that entered the 1.3 m detection corridor too late or not at all, a trade-off from narrowing the corridor to suppress false-positive braking; the rest involved agents from the side or rear-left (2), low-speed contacts where distance estimates stagnated (2), or braking too late or not to a complete stop (2).

Side effects of emergency braking. In the first rollout set, AEB increases the number of rear-ended scenes from 15 to 25. While these are not at-fault in the simulator, abrupt braking poses a genuine safety hazard in real-world traffic; moreover, 3 previously full-score scenes incurred new at-fault collisions under AEB.

Dominance of lateral failures. In the first AEB rollout, off-road departures and lateral corridor exits account for 0.253 of the 0.393 total score loss—failure modes that longitudinal AEB cannot mitigate. Off-road violations occurred at a median deviation of only 0.67 m from the map route (comparable to normal deviations in successful rollouts), so we opted against deploying a route-distance guard.

Suspected reliance on ego motion. Without AEB, first-incident front collisions occur in 25.7% of masked policy rollouts (77/300) vs. 29.9% for the baseline (90/301). The visual mask appears to reduce this failure mode, but the final plan commanded no deceleration in 64% vs. 68% of these collisions (Table 14).

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A Full configuration

Table 3 lists the training configuration and Table 4 the deployed driver and AEB configuration.

Table 3: Training configuration (identical for both runs unless otherwise stated). Source: W&B configurations of `ccb412c9` and `03bdfc50`; the model bundles deployed in the container images carry identical values.

Item	Value
Epochs / steps	12 / 38,400 (3,200 steps per epoch)
Hardware	1 Aurora node, 12 XPU tiles, one rank per tile (batch size 4 per tile)
Precision	bf16; gradient checkpointing enabled; max gradient norm 1
Optimizer	AdamW, betas $(0.9, 0.95)$, $\epsilon = 10^{-8}$, weight decay 0
Learning rates	VLM 10^{-6}; expert 5×10^{-5}; gate 10^{-4} (mask run)
Schedule	200 warmup steps, cosine decay to 10^{-8}; VLM frozen in epoch 1, then warmed up and decayed over the rest
Repeated diffusion steps	4 noise/time draws per image
Prompt	"predict trajectory" (no target or speed in the prompt)
Images	Front wide-angle camera; pixel budgets 921,600 (t_0) and 174,080 ($t_0 - 0.1\,\mathrm{s}$); 1 token per 32×32 pixels
VLM	`Qwen/Qwen-Drive-1.0-4B`: 32 layers, hidden dim 2560, SDPA attention
Expert	LayerwiseFM DiT: 32 blocks interleaving cross- and self-attention, width 1024, 16 heads of dim 64, dropout 0.2, AdaNorm
State	Horizon 5, dim 8: x, y, $\mathrm{heading}$, speed, a_{lon}, x_{tp}, y_{tp}, flag; per-dim q01/q99 scaled to $[-1, 1]$
Action	Horizon 25, dim 2: acceleration, curvature; per-dim z-score
Flow matching	Beta(1.5, 1) time, $s = 0.999$; velocity MSE; 10 Euler steps at inference; 1000 time buckets
Target dropout	$p = 0.5$ per sample (Path A in the mask run); target columns zeroed, flag set to 0
Eval	Every 250 steps on 240 fixed validation samples; open-loop
Mask (mask run only)	Scheme `three_path`; budgeted parameterization; $\rho = 0.2$; contrast 2; head hidden dim 256
Gate query	Target point $/25$ and flattened normalized ego state without target columns
Gate input	Final-layer VLM hidden states of visual tokens, detached, standardized per channel; RMSNorm

Table 4: Deployment configuration. Replanning, CUDA graph, and detector settings correspond to default parameters or pre-configured overrides baked into the submitted container images.

Item	Value
Replanning	Every 5th 100 ms interval (0.5 s); plan cached in the local frame and re-sliced at each tick
Temporal ensembling; action samples	Off (1 plan; 1 sample) across all three submissions
Linear-attention kernels	`auto`: flash-linear-attention Triton kernels, falling back to the reference implementation
Expert denoising	Replayed from a captured CUDA graph
Target point	25 m along the route; route resampled at 0.1 m and extended 2 m past the farthest target
Controller gains	Official defaults (no `-controller-gains` override)
Detector	RT-DETRv2-R18, COCO, 640×640 on the front wide-angle frame, score > 0.5, no NMS (DETR queries)
Classes	person, bicycle, car, motorcycle, bus, train, truck
Ground projection	7 points along the central 40% of the box bottom edge; flat ground $z = 0$ in the rig frame; range < 80 m
Corridor	1.3 m half-width around the planned path; path linearly extrapolated to 80 m
Front overhang	4.05 m (rear axle to front bumper); detections behind -1 m relative to the bumper ignored
Track	1 s window; at least 6 samples (10 Hz) or 3 (5 Hz); reset on a jump above 4 m or missed detection
Stop gap; latency	2.5 m; 0.4 s
Brake trigger; deceleration	$a_{\mathrm{req}} > 3$ m/s^2 or $g \leq 0$; $\mathrm{clip}(1.2\, a_{\mathrm{req}}, 3, 8)$ m/s^2
No-acceleration condition	$d/c < 3$ s
Re-timed plan grid	0.1 s
Detection rate	Every front frame (10 Hz variant); at least 0.2 s apart (5 Hz variant)

B Data and action representation

Sampling. We use all clips containing valid egomotion and front-camera streams from the dataset's training split. Each dataset item corresponds to a single clip with the anchor timestamp t_0 sampled uniformly from its valid range, ensuring that each epoch visits every clip once at a random timestamp. Only the densely sampled portion of the egomotion trajectory (spanning approximately 20 s around the camera window) is loaded. The two visual inputs are the front wide-angle frames closest to t_0 and $t_0 - 0.1$ s.

State. At each historical step $t \in \{t_0 - 0.4\,\mathrm{s}, \ldots, t_0\}$, the ego state consists of $[x, y, \mathrm{heading}_{\mathrm{rel}}, \mathrm{speed}, a_{\mathrm{lon}}]$ expressed in the ego coordinate frame at t_0. All of these quantities are directly provided by AlpaSim at deployment. The target point, also expressed in the ego frame at t_0, is repeated across all historical steps and followed by a binary presence flag.

Target point. The target point is defined as the first future ego position located at a Euclidean distance of at least 25 m from the current position, or the final recorded position of the clip if the trajectory terminates earlier. In the training run for our deployed model, this fallback endpoint case was retained (a subsequent experiment in Appendix H omits it, because a clip endpoint implicitly indicates where the ego vehicle stops—information that a map-based route target never provides at deployment).

Action. The action representation consists of 25 per-step [acceleration, curvature] pairs starting from t_0, extracted via Alpamayo 1.5's regularized least-squares fitting procedure (which jointly smooths heading, velocity, acceleration, and curvature, clamping curvature to 0 at speeds below 0.6 m/s) and integrated via the trapezoidal rule from the current speed v_0 to yield 2D poses at $t_0 + 0.1\,\mathrm{s}, \ldots, t_0 + 2.5\,\mathrm{s}$.

Statistics and validation. Feature normalization statistics are computed from 6,000 training clips $\times$ 8 time steps. The fixed validation set comprises 240 samples stratified into 80 samples per maneuver category, classified by the cumulative heading change over the subsequent 3 s using a $\pm 15^\circ$ threshold.

C Mask details and diagnostics

Budgeted mask. Given raw token scores ℓ_p within a single frame, we standardize them as $z_p = (\ell_p - \bar{\ell})/(\mathrm{std}(\ell) + 10^{-6})$ and compute soft mask values $m_p = \sigma(b + \alpha z_p)$ with $\alpha = 2$, where 40 bisection iterations over $b \in [-30, 30]$ enforce $\mathrm{mean}_p\, m_p = \rho$. The bisection solver itself is not differentiated directly. Instead, because the constraint $\sum_p \sigma(b + \alpha z_p) = N\rho$ defines $b(z)$ implicitly with $\partial b/\partial z_q = -\alpha w_q$ and $w = \sigma' / \sum \sigma'$, we add a zero-valued surrogate term $-\alpha \sum_q w_q (z_q - \mathrm{sg}[z_q])$ to supply this exact analytical gradient during backpropagation. By construction, a uniform

mask is precluded and the dispersion of m is fixed by α, preventing the mask from collapsing to an all-keep state or degenerating into a binary mask.

Where it acts. In every cross-attention block of the action expert, the attention logits are biased by adding $\log m_p$ to visual tokens, 0 to text tokens, and $-\infty$ to padding tokens. The VLM itself remains unmasked. A single shared gating head predicts masks for both input frames, enforcing the budget constraint independently per frame. Unlike S2 [16], which multiplies token embeddings by a floored mask inside the action executor, the mask here acts only through these attention logits.

Mask diagnostics. At step 38,400 the mask depends on the sample: the Spearman rank correlation between the masks of two samples is 0.055, and the mean row and column patterns explain 2.4% and 0.3% of its variance. The DiT's cross-attention largely follows the mask (rank correlation 0.827), and 26.1% of the attention mass falls on text tokens. The mask shows no clear lateral preference by maneuver: keep on the left minus the right image half is 0.002, 0.025, and 0.011 for left turns, right turns, and straight driving (Table 5).

Table 5: Mask diagnostics across the validation set (current frame) at step 38,400.

Diagnostic	Value
Mean keep, history frame	0.200
Mean keep, current frame	0.200
Spearman rank correlation between masks of two samples (1 = identical fixed pattern)	0.055
Mask variance explained by the mean row pattern	0.024
Mask variance explained by the mean column pattern	0.003
Spearman rank correlation between a sample's mask and its DiT attention	0.827
DiT cross-attention mass on text tokens	0.261
Keep on image left minus right half, left turns	0.002
Same, right turns	0.025
Same, straight	0.011

Gradients. Path A updates both the VLM and the DiT. Path B uses a detached mask and detached VLM features to update only the DiT. Path C uses detached VLM features, and the trainer applies its gradients exclusively to the gating module. The combined action loss is L_A+L_B, whereas L_C is backpropagated in a separate pass. All three pathways share the same sampled noise and flow time step, and losses L_B and L_C are weighted by whether the sample retained the target point in Path A.

Controls. *No mask*: identical inputs, omitting the $\log m$ bias. *Shuffled*: randomly permuting the predicted retention weights m_p across spatial positions within each frame. *Mirrored*: evaluating the gating module with a laterally mirrored target point ($y \to -y$). *No image tokens*: setting retention weights to 0 on every visual token. *Black frames*: replacing all camera inputs with black images and re-running the VLM without masking. *No target*: setting the target-presence flag to 0 and zeroing the target columns in the DiT state; when combined with the mask, target information reaches the policy solely through the gating module.

Training signals. The flow-matching loss of Path A falls from 0.96 to 0.18 and ends level with the baseline's (Table 6). Mean keep stays at 0.200 by construction. Path C ends at 0.0122, below its shuffled-mask (0.0160) and no-mask (0.0156) controls, so the learned mask does make Path C closer to Path A when the DiT receives no target; these controls carry no gradient. The step time rises from 2.3 s to 7.1 s and throughput falls accordingly once the VLM is unfrozen after the first epoch. For the baseline, the curvature loss ends well below the acceleration loss (0.096 vs. 0.264). Fig. 4 shows the same signals over training.

Table 6: Training signals: mean of the first 50 and last 200 logged points (logged every 10 steps).

	Mask run		Baseline	
Signal	First	Last	First	Last
Flow-matching loss (Path A)	0.9586	0.1790	0.9424	0.1796
Loss, acceleration	–	–	0.9591	0.2635
Loss, curvature	–	–	0.9256	0.0957
Path B (distillation, DiT)	0.0521	0.0071	–	–
Path C (distillation, gate)	0.0674	0.0122	–	–
Path C, shuffled mask	0.0688	0.0160	–	–
Path C, no mask	0.0680	0.0156	–	–
Mean keep, current frame	0.2000	0.2000	–	–
Mean keep, history frame	0.2000	0.2000	–	–
Std of gate scores	2.0426	3.4981	–	–
Fraction with target (Path A)	0.4987	0.5026	0.4944	0.4997
Gradient norm	24.555	4.1008	24.700	3.7741
Samples per second	20.943	6.6069	28.595	6.9585
Step time (s)	2.2586	7.1135	–	–

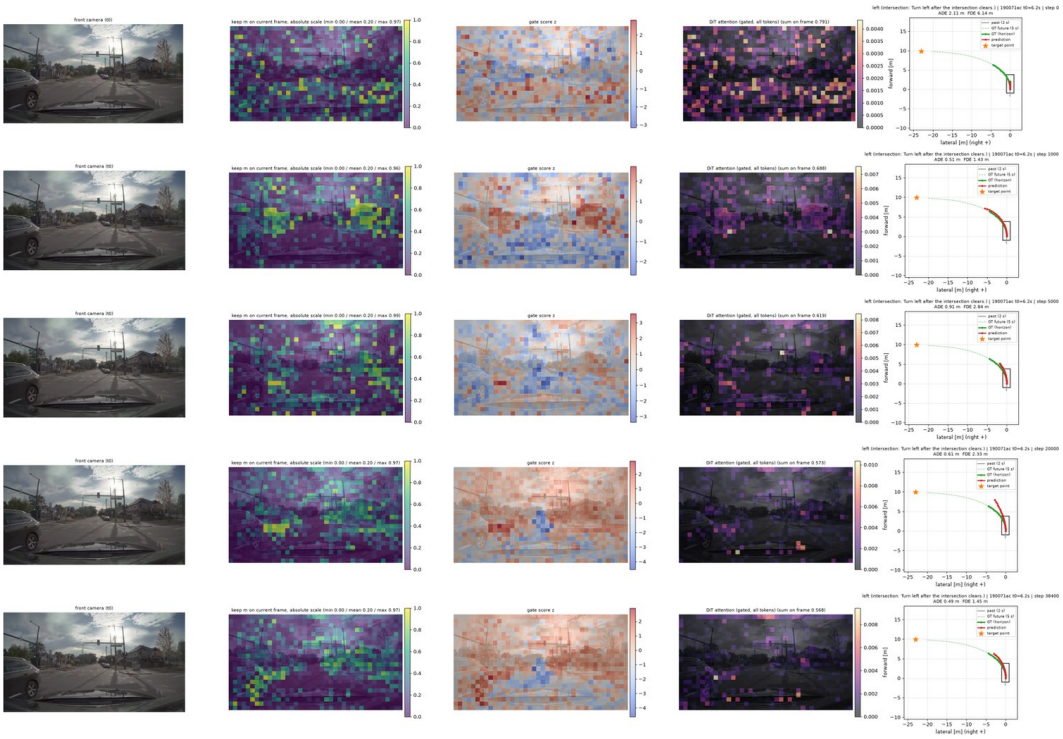


Figure 2: Evolution of the visual mask on a left-turn validation sample across training: steps 0, 1,000, 5,000, 20,000, and 38,400 (top to bottom). Panels from left to right: current front-camera frame; retention weight m; standardized gate score z; DiT cross-attention map on the frame; and bird's-eye view showing past trajectory, ground truth, prediction, and target point (lateral axis positive to the right).

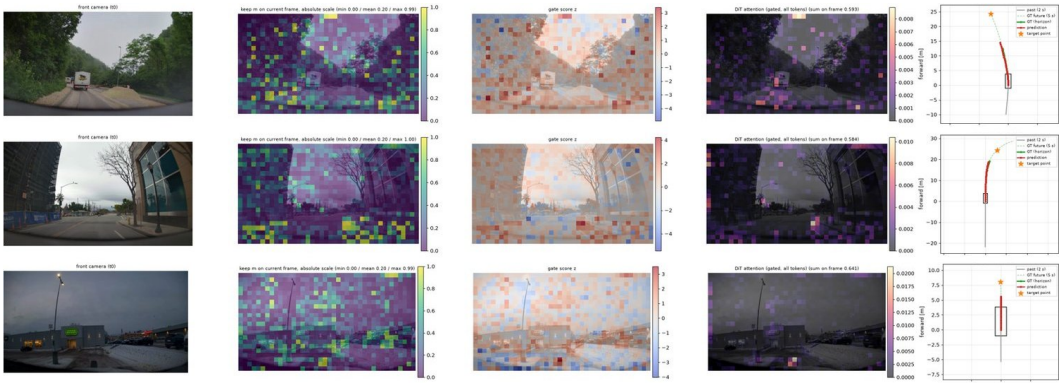


Figure 3: Mask-run validation samples at step 38,400: a left turn, a right turn, and straight driving (top to bottom), with panels arranged as in Fig. 2.

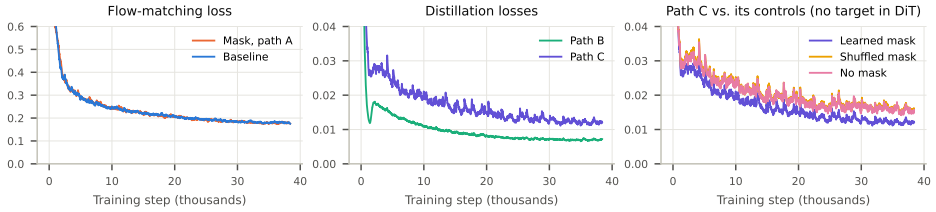


Figure 4: Smoothed training losses (EMA factor 0.05). The baseline flow-matching loss closely tracks Path A of the masked run, while Path C with the learned mask remains consistently below its shuffled-mask and unmasked controls.

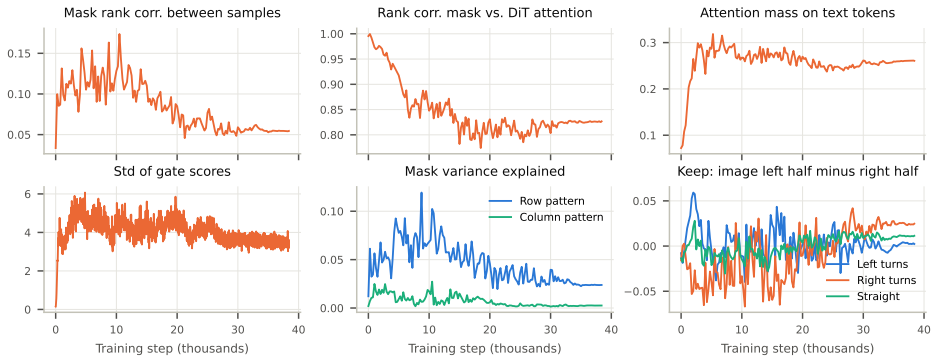


Figure 5: Mask diagnostics across training. The rank correlation between the mask and DiT cross-attention decreases from near 1 to approximately 0.83, cross-attention mass on text tokens increases early in training, and mean row/column spatial patterns account for minimal mask variance.

D Open-loop results in full

Both models use the final training checkpoint at step 38,400. On the fixed 240-sample validation set, the lowest validation error observed during training is slightly lower than that of the final checkpoint: the baseline achieved a minimum ADE of 0.302 at step 32,250 (vs. 0.337 final), and the masked policy achieved 0.324 at step 34,750 (vs. 0.341 final). We did not perform checkpoint selection on this validation set. Because each open-loop evaluation draws a single flow-matching trajectory sample per input, these metrics include stochastic sampling variance; no confidence intervals were computed.

Table 7: Open-loop metrics at step 38,400 on the 240 fixed validation samples (in meters; miss rate as a fraction). L2 denotes position error at the specified horizon (in seconds); Lat./Lon. denote absolute final lateral and longitudinal errors.

	Run	ADE	FDE	L2@0.5	L2@1.0	L2@1.5	L2@2.0	L2@2.5	Lat.	Lon.	Miss
All	Baseline	0.337	1.029	0.025	0.117	0.300	0.574	0.943	0.478	0.802	0.125
	Mask	0.341	1.023	0.028	0.126	0.308	0.578	0.939	0.457	0.796	0.125
Left	Baseline	0.405	1.213	0.034	0.151	0.365	0.686	1.115	0.768	0.790	0.200
	Mask	0.376	1.100	0.034	0.146	0.347	0.637	1.014	0.667	0.712	0.150
Right	Baseline	0.367	1.136	0.025	0.117	0.320	0.633	1.041	0.516	0.931	0.087
	Mask	0.360	1.089	0.031	0.136	0.321	0.606	0.997	0.526	0.867	0.100
Straight	Baseline	0.239	0.737	0.017	0.083	0.215	0.404	0.672	0.149	0.686	0.087
	Mask	0.287	0.879	0.020	0.097	0.255	0.490	0.806	0.179	0.810	0.125

Table 8: Diagnostic controls for the masked policy at step 38,400, sharing the sampled noise of the main prediction. Each three-column group reports ADE, FDE, and miss rate.

	All			Left			Right			Straight		
Inference path	0.341	1.023	0.125	0.376	1.100	0.150	0.360	1.089	0.100	0.287	0.879	0.125
No mask	0.339	1.016	0.129	0.376	1.096	0.138	0.358	1.086	0.125	0.282	0.866	0.125
Shuffled mask	0.344	1.031	0.133	0.382	1.115	0.150	0.363	1.099	0.125	0.287	0.879	0.125
Mirrored target in gate	0.348	1.049	0.129	0.392	1.154	0.175	0.363	1.114	0.087	0.288	0.881	0.125
No target in DiT, no mask	0.367	1.135	0.167	0.395	1.217	0.188	0.403	1.261	0.163	0.303	0.926	0.150
No target in DiT, mask	0.359	1.094	0.142	0.383	1.150	0.163	0.391	1.210	0.125	0.302	0.921	0.138
No target in DiT, shuffled mask	0.374	1.155	0.163	0.406	1.249	0.175	0.411	1.285	0.175	0.305	0.932	0.138
All image tokens removed	0.456	1.418	0.242	0.469	1.449	0.263	0.507	1.601	0.275	0.393	1.204	0.188
Black input frames	0.486	1.564	0.304	0.518	1.598	0.362	0.561	1.882	0.375	0.379	1.211	0.175
Baseline, black frames	0.441	1.408	0.246	0.528	1.602	0.300	0.476	1.568	0.287	0.318	1.052	0.150

Table 9: Turn response metrics at step 38,400. Direction accuracy and lateral displacement ratios are computed over left/right turn samples with $|y_{\mathrm{GT}}| > 0.5\,\mathrm{m}$.

	Baseline	Mask	No mask	Shuffled	Mirrored	No target	No img tokens	Black	Base. black
Corr. of final lateral offset	0.986	0.987	0.987	0.987	0.986	0.984	0.967	0.959	0.969
Std of predicted final lateral (m)	4.451	4.482	4.488	4.497	4.384	4.475	3.981	3.668	3.993
Left: direction accuracy	0.987	1.000	1.000	1.000	0.987	1.000	1.000	1.000	0.987
Right: direction accuracy	1.000	1.000	1.000	1.000	1.000	0.987	0.973	1.000	0.973
Left: median pred./GT lateral	0.993	1.002	1.000	0.993	0.961	0.984	0.919	0.897	0.931
Right: median pred./GT lateral	1.031	1.020	1.018	1.025	0.973	0.986	0.823	0.726	0.838

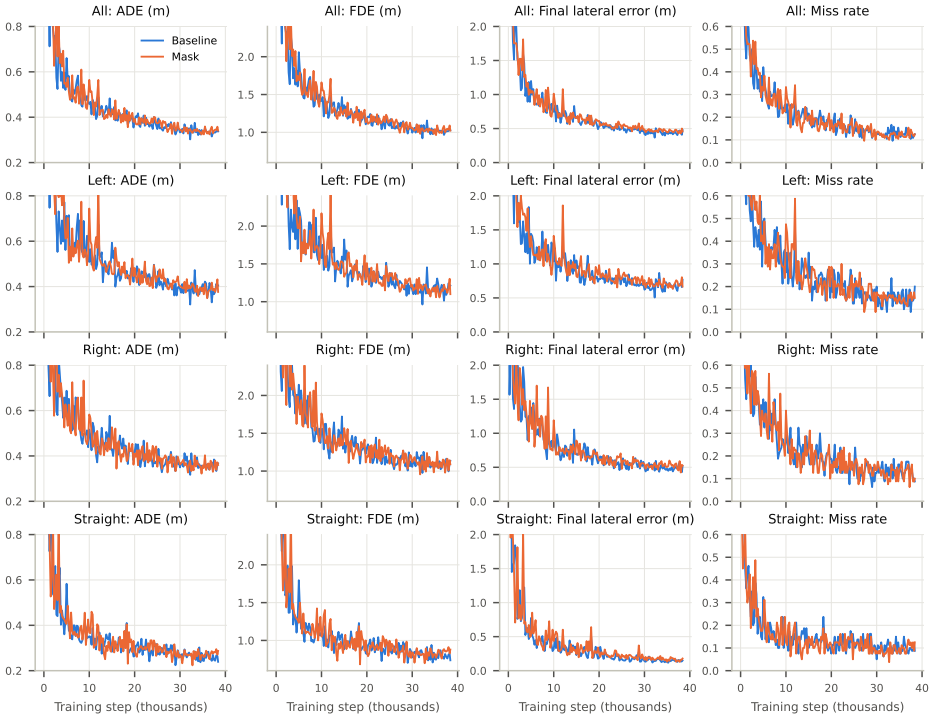


Figure 6: Open-loop validation metrics evaluated every 250 training steps. Vertical axes are clipped for readability, placing initial early-training evaluations off-scale.

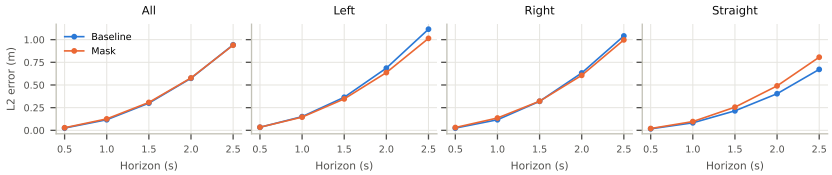


Figure 7: L2 position error across the 2.5 s prediction horizon at step 38,400.

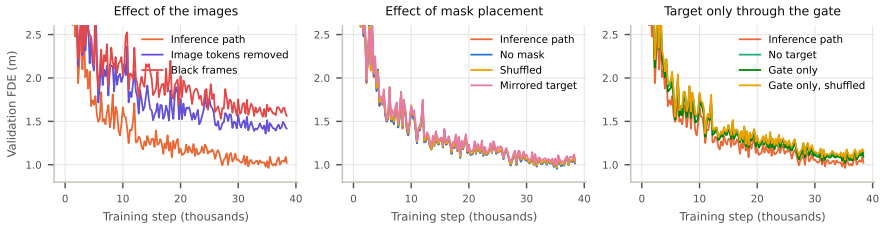


Figure 8: Mask-run diagnostic controls across training (validation FDE). Control definitions are provided in Appendix C.

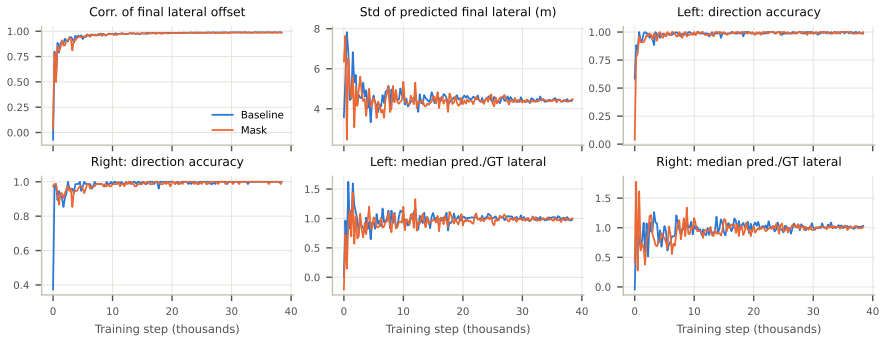


Figure 9: Turn response metrics across training.

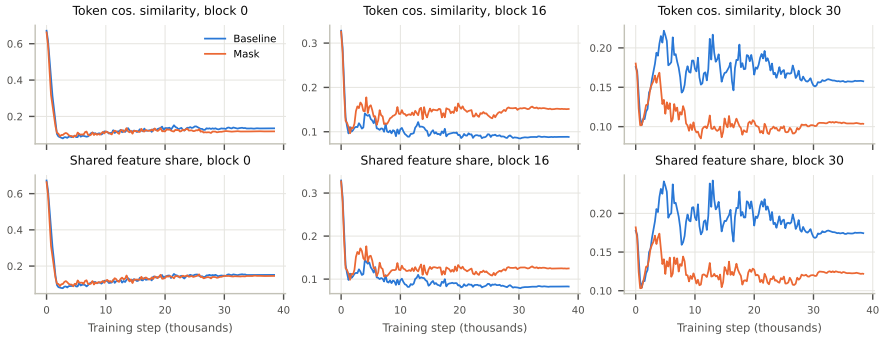


Figure 10: Cosine similarity among projected VLM tokens entering the DiT at three depth blocks. In the deepest block, token representations in the masked run exhibit lower pairwise similarity than in the baseline.

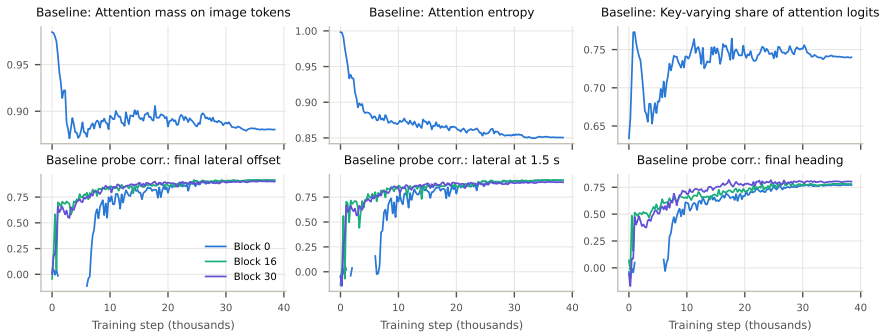


Figure 11: Baseline diagnostics: DiT cross-attention statistics (top), and correlations between detached probes (attention pooling and an MLP) on projected VLM features and future motion across three DiT blocks (bottom).

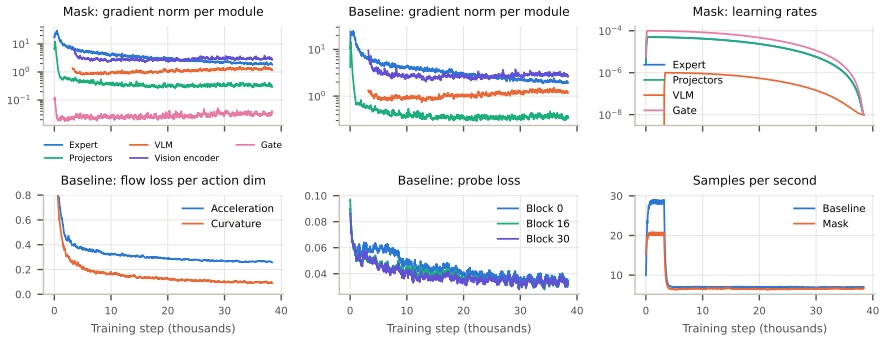


Figure 12: Smoothed optimization signals across training. Throughput is substantially higher during the first epoch while the VLM backbone is frozen.

E Driver pipeline

Observation handling. The driver service implements AlpaSim's gRPC API. Incoming JPEG-compressed camera frames are decoded upon arrival, with only the front wide-angle camera feeding the policy and object detector. Camera calibration is derived from the session's nominal f-θ intrinsics (scaled to the decoded frame resolution), falling back to the static rig configuration file if unavailable. Egomotion states are recorded into the latest observation bundle at every `Drive` call, ensuring that the 0.5 s replanning cycle always reads a complete 10 Hz state history.

Route. The challenge evaluation presets use map-based routes with a 40 m start offset. Whenever the route's initial waypoint lies beyond the resampling step, a cubic Hermite spline departing the rig origin along $+x$ and joining the route's first segment bridges the gap; the resulting polyline is then linearly extended beyond the farthest target lookahead distance and uniformly resampled at 0.1 m intervals. The target point is extracted exactly as in training, substituting the resampled map route for the logged future trajectory.

Serving. Each generated plan is anchored in the local coordinate frame at creation time. Every control tick emits the planned waypoints corresponding to timestamps at or after the current time on its 10 Hz grid. If a new plan cannot be computed (due to a missing camera frame, missing ego pose, or runtime planning exception), the previously cached plan remains active; if no valid cached plan exists, the driver falls back to serving a straight-line trajectory at constant speed $\max(2\,\mathrm{m/s}, v)$.

AEB at a tick. At each control tick, the AEB module executes five steps: (1) If the newest front-camera frame has not yet been processed and a detection is scheduled, run the detector and project the bounding boxes onto the ground plane. (2) Transform the remainder of the cached plan into the current rig frame and identify the nearest lead obstacle along it. (3) Update the obstacle tracker and compute the longitudinal speed cap. (4) When a speed cap is active, let $w(\tau)$ denote the distance that the cached plan still intends to cover over τ seconds, and let $u(\tau)$ denote the maximum distance permitted by the deceleration ramp (or $v\tau$ under the no-acceleration condition). The served trajectory starts at the ego vehicle's orthogonal projection onto the planned path, advancing to arc-length positions $s_{\mathrm{ego}} + \min(w, u)$ on a 0.1 s grid with yaw angles interpolated along the geometric path. (5) If the speed cap has just been released on the current tick, hold the current speed for one tick and invalidate the cached plan so that the policy replans on the next tick. Without Step (5), reverting directly to the cached plan—whose timestamped waypoints now lie far ahead of the braked ego vehicle—would cause the tracking controller to accelerate aggressively toward the obstacle it had just braked for.

F AEB development

Why it was added. Without AEB, the single largest source of score loss is at-fault collisions—predominantly front-end collisions with lead vehicles traveling substantially slower than the ego vehicle. Across 178 first-incident front collisions pooled from five evaluation runs (mask, TE4, hist_ft, and two sig02 rollouts), the median ego impact speed is 11.9 m/s (interquartile range: 6.8–17.8 m/s). Kinematic profiles over the 3 s preceding impact differ only moderately from non-incident 3 s windows prior to any incident (6,077 windows): a speed increase exceeding 2 m/s occurs in 35% vs. 25% of windows; mean acceleration exceeding 0.3 m/s^2 occurs in 48% vs. 37%; and peak acceleration exceeding 1.5 m/s^2 occurs in 24% vs. 21%. Imposing a global acceleration cap would therefore mostly cost route progress, motivating an obstacle-aware braking rule. Because the driver receives no privileged state for other traffic agents and ground-truth perception labels are disabled in the challenge presets, obstacles must be detected directly from camera images.

Offline replay. Each evaluation rollout log stores the front-camera JPEGs, ego states, and served trajectory plans. We executed the detector across these logged frames inside the driver container, cached the raw bounding boxes, and tuned the AEB decision rules via offline replay. An initial configuration (no bounding-box bottom-edge shrinkage, 1.6 m corridor half-width, 2 m/s^2 trigger threshold) triggered braking at least 1 s prior to impact in 38 of 41 front collisions, but also triggered false-positive braking in 29 of 46 sampled clean rollouts. Visual inspection traced these false positives primarily to cars parked along the curb: when viewed obliquely, the bottom corners of a vehicle's 2D bounding box extend laterally beyond the physical wheels and intrude into the planned

driving corridor. Table 10 illustrates the resulting trade-off across hyperparameter settings. Note that this offline replay is open-loop and tuned on the local evaluation scenes.

Table 10: Offline AEB hyperparameter sweep on replayed detections from the masked policy's 150 rollouts (comprising 41 front collisions and 62 incident-free rollouts). A braking event is counted only when the speed cap shortens the planned path at a 1 s horizon by more than 0.3 m. The selected deployment configuration is highlighted in bold. Evaluation is open-loop (ego motion does not react to simulated braking).

Shrink	Half-width	Trigger	Gap	Samples	Hits, ≥ 1 s	Hits, ≥ 0.5 s	Clean: rollouts / frames
0.0	1.6	3.0	2.5	4	36	38	38 / 769
0.3	1.3	4.0	1.5	6	24	28	15 / 92
0.3	**1.3**	**3.0**	**2.5**	**6**	**29**	**32**	**19 / 164**
0.3	1.3	4.0	2.5	6	29	32	17 / 127
0.15	1.3	3.0	2.5	6	32	37	22 / 227
0.15	1.45	3.0	2.5	6	34	38	24 / 268
0.3	1.6	3.0	2.5	6	31	36	22 / 214
0.0	1.3	4.0	1.5	6	28	34	24 / 234
0.3	1.3	2.0	2.5	6	31	32	23 / 239

Chosen configuration. Under the selected parameter setting, the 3 s time-to-collision no-acceleration rule introduces 47 additional frames of intervention across clean rollouts. In the AlpaSim scoring metric, being rear-ended by a trailing agent does not zero the scene score (as only front and lateral impacts are penalized as at-fault) and route progress saturates at 80% of the recorded reference distance; consequently, the challenge metric rewards this conservative braking trade-off. However, as noted in Section 7, inducing rear-end collisions would pose a severe safety hazard in real-world driving.

Detector choice. RT-DETRv2 is natively supported in our container image's `transformers` library with Apache-2.0 licensed weights, and its DETR-style set prediction architecture requires no post-processing non-maximum suppression (NMS). Profiling the R18 and R50 backbones inside the container on a heavily loaded NVIDIA RTX PRO 6000 Blackwell GPU yielded median per-frame latencies of approximately 38 ms and 49 ms, with peak VRAM allocations of 183 MiB and 298 MiB, respectively. Offloading image resizing to the CPU and enabling `fp16` autocast did not substantially alter R18 latency (approximately 35 ms), and processing a batch of 4 frames required nearly the same wall-clock time as a batch of 1 (approximately 30 ms), indicating that CPU-side kernel launch overhead dominated under multi-tenant host contention. Attempting to capture the detector within a CUDA graph failed because the upstream implementation performs CPU-to-GPU tensor writes during the `forward` pass.

G Closed-loop results in full

Protocol. Evaluations were conducted on a fixed 150-scene subset of the 441-scene `nurec_curated_val` benchmark using the `+e2e_challenge=dev` preset, configured with a multi-GPU topology and driver video recording enabled on shared multi-tenant hosts. The first rollout sets (baseline, mask, and 10 Hz AEB), as well as the TE4, hist_ft, and 8-sample variants, were executed on an NVIDIA RTX PRO 6000 Blackwell host (8 GPUs). The second rollout sets, along with the sig02 and 5 Hz AEB variants, were executed on NVIDIA RTX 6000 Ada hosts (3 GPUs each). Because the first and second rollouts for each configuration ran on distinct hardware architectures, discrepancies between the two rollouts reflect both hardware differences and stochastic run-to-run variance. Any scenes that terminated prematurely due to renderer out-of-memory errors on shared GPUs were re-run with reduced concurrency per renderer instance. Scoring terminates at the first collision or off-road departure: a rollout receives a score of 0 if it incurs an at-fault collision (front or lateral impact), an off-road violation, or a lateral corridor exit (deviating by more than 4 m from the recorded reference path), and $\min(\text{progress}/0.8, 1)$ otherwise.

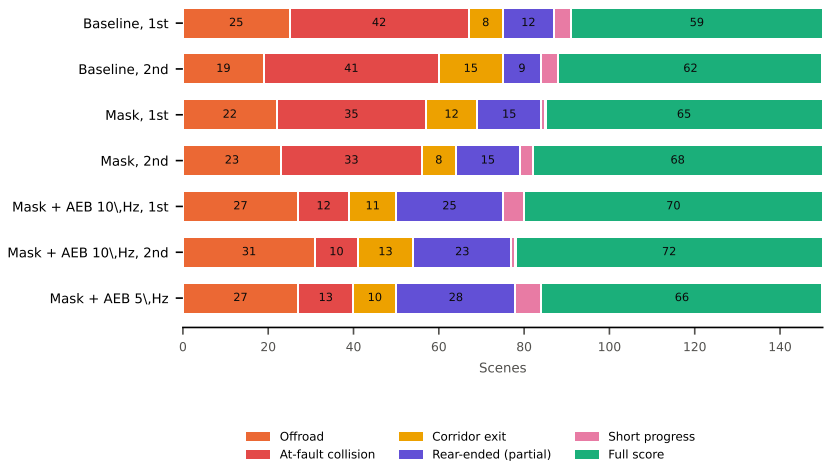


Figure 13: Per-scene rollout outcomes across the same 150 evaluation scenes.

Table 11: Score-loss decomposition across the same 150 scenes in every column, formatted as: affected scenes / contribution to (1 − mean score).

	Baseline, 1st	Baseline, 2nd	Mask, 1st	Mask, 2nd	AEB 10 Hz, 1st	AEB 10 Hz, 2nd	AEB 5 Hz
Offroad	25 / 0.167	19 / 0.127	22 / 0.147	23 / 0.153	27 / 0.180	31 / 0.207	27 / 0.180
At-fault collision	42 / 0.280	41 / 0.273	35 / 0.233	33 / 0.220	12 / 0.080	10 / 0.067	13 / 0.087
Corridor exit	8 / 0.053	15 / 0.100	12 / 0.080	8 / 0.053	11 / 0.073	13 / 0.087	10 / 0.067
Rear-ended (partial)	12 / 0.022	9 / 0.015	15 / 0.038	15 / 0.037	25 / 0.054	23 / 0.046	28 / 0.067
Short progress	4 / 0.002	4 / 0.003	1 / 0.002	3 / 0.002	5 / 0.005	1 / 0.002	6 / 0.009
Full score	59 / 0.000	62 / 0.000	65 / 0.000	68 / 0.000	70 / 0.000	72 / 0.000	66 / 0.000
Mean score	0.476	0.481	0.499	0.534	0.607	0.591	0.590
Mean progress	0.745	0.773	0.737	0.739	0.737	0.732	0.720
Mean Drive (s)	0.133	0.066	0.110	0.146	0.201	0.085	0.264
Host	Blackwell	Ada	Blackwell	Ada	Blackwell	Ada	Ada

Table 12: Paired per-scene comparisons across the same 150 scenes. CI denotes the 95% bootstrap confidence interval (10,000 resamples); p denotes the two-sided sign test p-value (ignoring ties, uncorrected for multiple comparisons); (2) indicates the per-scene mean across two rollouts.

Comparison	n	Ref.	Run	Diff.	95% CI	W/L	p
Mask (2) vs. baseline (2)	150	0.479	0.517	+0.038	[-0.018, +0.093]	37/26	0.21
Mask 1st vs. baseline 1st	150	0.476	0.499	+0.023	[-0.049, +0.093]	28/20	0.31
Mask 2nd vs. baseline 2nd	150	0.481	0.534	+0.053	[-0.022, +0.126]	31/19	0.12
Mask 2nd vs. mask 1st	150	0.499	0.534	+0.035	[-0.017, +0.089]	22/16	0.42
Baseline 2nd vs. baseline 1st	150	0.476	0.481	+0.005	[-0.065, +0.076]	25/20	0.55
AEB 10 Hz (2) vs. mask (2)	150	0.517	0.599	+0.082	[+0.023, +0.143]	41/28	0.15
AEB 10 Hz 1st vs. mask 1st	150	0.499	0.607	+0.108	[+0.038, +0.176]	37/14	<0.01
AEB 10 Hz 2nd vs. mask 2nd	150	0.534	0.591	+0.057	[-0.024, +0.137]	34/26	0.37
AEB 10 Hz 2nd vs. 1st	150	0.607	0.591	-0.016	[-0.087, +0.054]	24/30	0.50
AEB 5 Hz vs. mask (2)	150	0.517	0.590	+0.073	[+0.012, +0.134]	43/22	0.01
AEB 5 Hz vs. mask 2nd	150	0.534	0.590	+0.056	[-0.013, +0.125]	36/21	0.06
AEB 5 Hz vs. AEB 10 Hz (2)	150	0.599	0.590	-0.009	[-0.065, +0.044]	34/34	1.00

Where the collisions go. Table 13 follows each scene through the first rollouts on the same host. Of the 35 scenes with an at-fault collision without AEB, 9 remain at-fault collisions, 9 reach the full-score outcome, and 11 become rear-ended scenes, while 4 turn into lateral failures (2 off-road, 2 corridor exits) and 2 end with short progress. In the other direction, of the 65 full-score scenes without AEB, 56 stay at full score, 3 gain an at-fault collision, 4

Table 13: Scene outcome transition matrix when adding 10 Hz AEB (first rollouts, same host, 150 scenes): rows denote outcomes without AEB, and columns denote outcomes with AEB.

Without \ with	Offroad	Coll.	Exit	Rear	Short	Full
Offroad	19		1			2
At-fault collision	2	9	2	11	2	9
Corridor exit	3		5	1		3
Rear-ended (partial)	1		1	11	2	
Short progress					1	
Full score	2	3	2	2		56

gain a lateral failure, and 2 are rear-ended. Scenes that were already off-road without AEB mostly stay so (19 of 22).

Collision kinematics without AEB. Table 14 compares pre-collision kinematic statistics across model variants. Among the first-incident front collisions without AEB, the ego vehicle accelerated by more than 2 m/s over the preceding 3 s in 30% (mask) vs. 42% (baseline) of cases, and the final planned trajectory failed to command deceleration in 64% vs. 68% of cases.

Table 14: Pre-impact kinematics for first-incident front collisions across the same 150 scenes: overall front-collision rate; fraction with a speed increase > 2 m/s over the preceding 3 s; fraction whose final planned trajectory failed to command deceleration (planned speed 2 s ahead $\geq v - 0.3$ m/s); fraction where the nearest lead actor was traveling below 1 m/s (approximate); and median ego impact speed (m/s).

	Hits	Rate	Accel.	No slowing	Stopped lead	v_{hit}
Baseline (2)	90/301	29.9%	42%	68%	39%	11.6
Mask (2)	77/300	25.7%	30%	64%	38%	10.9
Mask + AEB 10 Hz (2)	28/300	9.3%	32%	50%	61%	9.1
Mask + AEB 5 Hz	12/150	8.0%	17%	42%	67%	2.4
hist_ft	46/147	31.3%	46%	67%	39%	13.2
sig02	50/150	33.3%	32%	62%	36%	12.9
Mask + TE4	41/150	27.3%	29%	66%	34%	10.5
Baseline + TE4	34/148	23.0%	35%	62%	44%	9.9

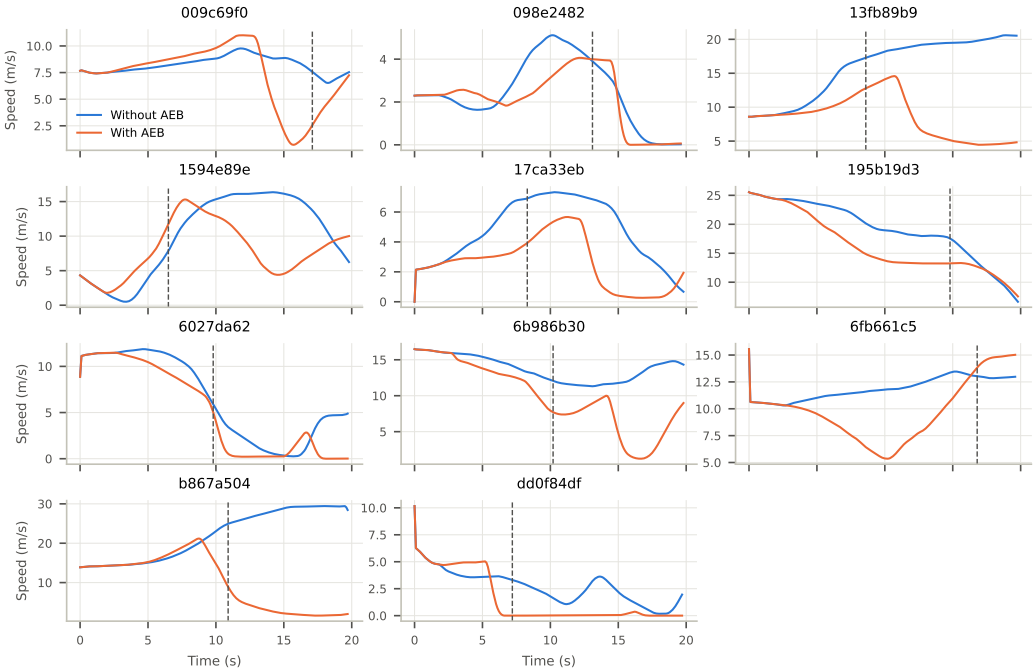


Figure 14: Ego speed profiles across the 11 scenes that incurred an at-fault collision without AEB but achieved a full score with 10 Hz AEB enabled (first rollouts). Dashed vertical lines mark the time of collision without AEB.

Remaining collisions with AEB (first rollout). Table 15 details all 12 remaining at-fault collisions under 10 Hz AEB. *Corridor*: the obstacle remained outside the 1.3 m lateral detection corridor until shortly before impact (or throughout the approach); relaxing the geometry to zero edge shrinkage and a 2 m half-width in offline replay detects most of these obstacles earlier, but at the cost of the false-positive braking events analyzed in Table 10. *Side/behind*: the colliding agent approached from outside the forward corridor (e.g., a crossing bus or a vehicle overtaking from the rear-left). *Close range*: the ground-projected distance estimate stagnated near 4.4 m, preventing the linear regression tracker from observing a positive closing speed; we hypothesize (though have not formally verified) that the ego vehicle's hood occluded the lead vehicle's wheels at close range. *Late brake*: in scene `1316f5fa`, a step discontinuity in estimated distance reset the obstacle track, delaying braking until approximately 0.5 s before impact; in scene `382a7247`, AEB decelerated the ego vehicle from 9.2 m/s down to 0.7 m/s, but low-speed contact still occurred.

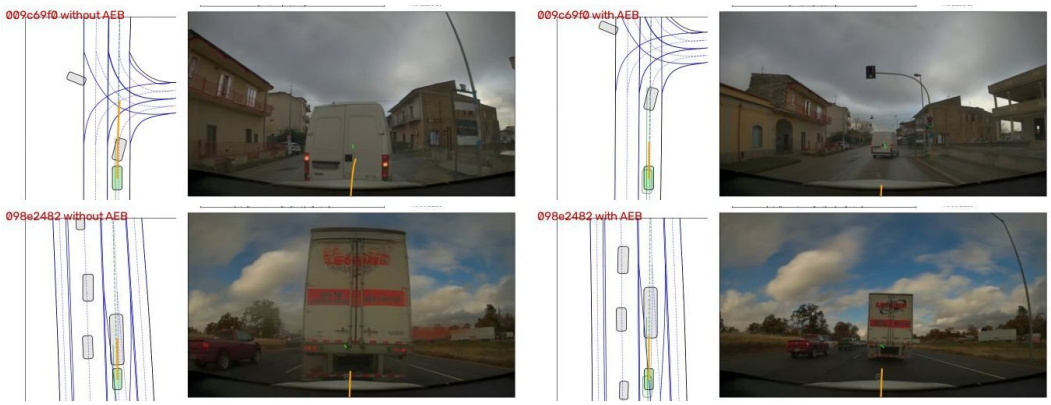


Figure 15: Side-by-side comparison 0.3 s prior to collision without AEB versus the same elapsed rollout time with AEB enabled, for 2 of the 11 rescued scenes. Each tile shows the map view (left) and the camera view (right).

Table 15: Breakdown of the 12 remaining at-fault collisions under 10 Hz AEB (first rollout). Columns report: elapsed time t from rollout start (s); ego speeds 3 s prior (v_{-3}) and at impact (v) in m/s; minimum acceleration $a_{\min}$ over the last 3 s (m/s^2); kinematics of the nearest traffic agent at impact (speed v_o, longitudinal and lateral center offsets in the ego frame, and relative heading in degrees; approximate, as the nearest agent may not be the struck actor); and lead time (s) prior to impact at which the AEB trigger condition was first met in offline detector replay. Failure categories reflect manual inspection.

Scene	Class	Type	t	v_{-3}	v	$a_{\min}$	v_o	Lon.	Lat.	Head.	Brake
0caa8f1a	Close range	front	9.4	0.1	1.2	-1.5	0.5	4.4	-1.4	-10	4.5
d641893f	Close range	front	15.1	0.9	4.2	0.8	12.0	-1.1	3.1	-179	none
05855d57	Corridor	front	11.5	30.3	32.0	-0.7	26.8	4.3	-0.0	2	0.3
213dfdac	Corridor	front	3.2	11.3	8.7	-1.5	0.1	2.2	-1.9	7	none
24a50fcc	Corridor	front	11.3	9.8	9.7	-0.7	7.1	1.9	-1.9	4	none
8b25a6ac	Corridor	side	10.6	9.9	9.9	-0.1	0.0	2.7	-2.1	-1	none
d5f0f729	Corridor	front	18.2	24.3	21.7	-1.5	0.1	4.2	-1.6	1	none
fadc73da	Corridor	front	17.1	1.4	6.5	-0.2	0.0	3.7	2.1	4	none
1316f5fa	Late brake	front	19.8	6.6	7.0	-8.3	0.1	4.3	-1.7	-5	0.5
382a7247	Late brake	front	9.0	9.2	0.7	-8.6	0.0	4.0	1.8	-162	2.9
0d79a6a0	Side/behind	side	3.2	10.5	8.1	-5.5	11.8	0.1	2.5	-22	0.9
1d879259	Side/behind	front	6.0	2.1	4.6	-2.9	6.0	4.6	-0.8	-71	0.3

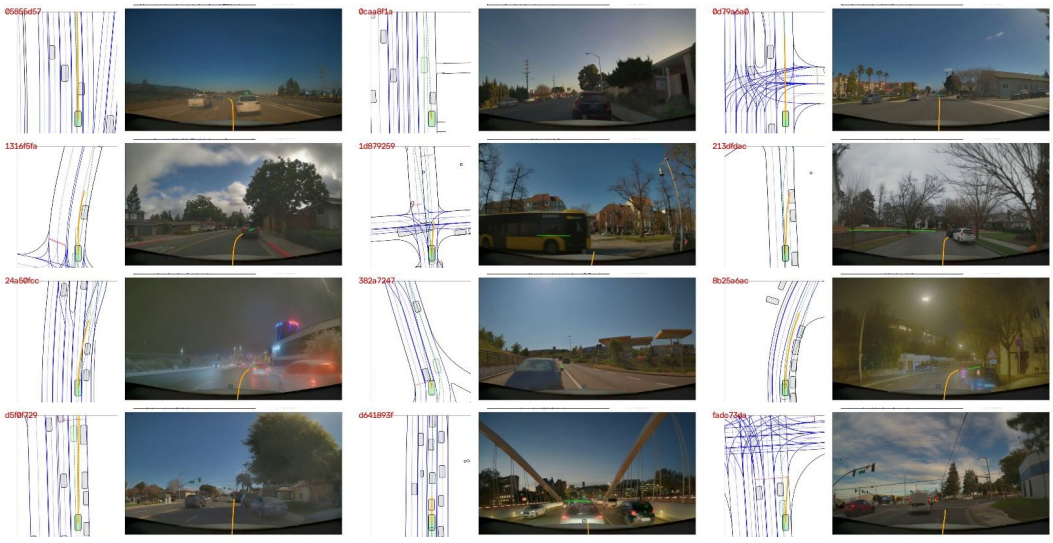


Figure 16: Views 1 s prior to impact for all 12 remaining at-fault collisions under 10 Hz AEB (first rollout); Map view (left) and camera view (right); Table 15 classifies them.

Lateral failures with AEB (first rollout). Among the 27 scenes resulting in off-road violations under AEB, 19 also went off-road in the first rollout without AEB, and only 3 experienced hard braking (deceleration below -2 m/s^2) during the preceding 3 s, indicating that AEB intervention is not the primary cause of off-road departures. Eighteen of the 27 scenes exhibited a cumulative heading change of less than 15° over the final 3 s, consistent with gradual lateral drift, and video

inspection of 12 sampled scenes frequently shows the planned trajectory drifting toward the right shoulder. Among the 31 lateral failures (off-road or corridor exit) with a well-defined lateral offset direction, 17 occurred to the right of the route and 9 to the left, while 5 occurred within 0.3 m of the route centerline.

Route as a lateral reference. Because provided routes are generated from map lane centerlines, they represent a natural candidate reference for lateral safety guarding. When sampling route waypoints every 2 m and placing the ego vehicle footprint along the route, only 114 of 36,843 sampled poses (0.3%) trigger an off-road violation under the simulator's scoring logic (5,728 poses exceed a single lane's boundaries without intersecting a road edge). Thus, while the map route lies almost entirely within drivable road boundaries, Tables 16 and 17 demonstrate that lateral distance to the route poorly discriminates impending lateral failures from normal driving variations; consequently, we did not deploy a route-distance guard.

Table 16: Offline evaluation of a hypothetical route-distance guard (not deployed): number of lateral failure rollouts flagged at least 1 s prior to failure versus number of full-score clean rollouts falsely flagged at any point.

	Ego		Plan at 2 s	
Thr. (m)	Fail.	Clean	Fail.	Clean
0.5	27/38	50/70	29/38	49/70
0.75	21/38	37/70	26/38	39/70
1.0	19/38	30/70	23/38	33/70
1.5	15/38	17/70	17/38	20/70
2.0	13/38	15/70	15/38	16/70
3.0	8/38	7/70	12/38	9/70

Table 17: Lateral distance (m) from the ego vehicle to the served map route prior to lateral failures under 10 Hz AEB. Scenes lacking a covering route segment at the queried timestamp are omitted. For reference, across all frames of the 70 full-score rollouts, the lateral distance distribution has a median of 0.29 m, 90th percentile of 1.14 m, and 99th percentile of 2.99 m.

Failure	Before	n	Median	p25	p75	> 1 m	> 2 m
Offroad	-3 s	18	0.40	0.18	2.56	5	5
	-2 s	23	0.49	0.31	1.36	6	4
	-1 s	23	0.70	0.33	1.91	11	5
	-0 s	23	0.67	0.40	1.99	8	6
Corridor exit	-3 s	5	0.81	0.70	0.87	1	1
	-2 s	7	0.79	0.34	2.77	3	3
	-1 s	8	0.82	0.26	2.36	4	3
	-0 s	8	2.80	0.61	3.43	5	5

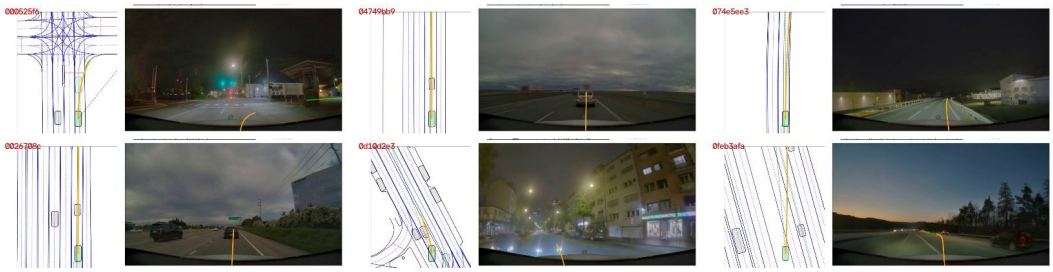


Figure 17: Views 0.8 s prior to the event under 10 Hz AEB (first rollout), in scene-ID order: the first 3 of the 27 off-road failures (top) and the first 3 of the 11 lateral corridor exits (bottom). Map view (left) and camera view (right) of each scene.

H Variants evaluated and not submitted

All additional variants were evaluated using the same local closed-loop protocol without AEB. We report these experiments as negative or inconclusive results:

- **Temporal ensembling (TE4).** Following the temporal ensembling scheme of ACT [18], each newly predicted trajectory plan is averaged with up to 3 preceding plans across their overlapping time horizons; each older plan is rigidly transformed so that its pose at the new plan's creation timestamp aligns with the current ego pose, using exponential decay weights $\exp(-0.5 \cdot \text{age})$.
- **8 samples.** Averaging 8 independent flow-matching trajectory samples per planning call within the normalized action space.
- **hist_ft.** A 3-epoch masked fine-tuning run initialized from the baseline's final checkpoint, incorporating ego-history dropout ($p = 0.5$) with a binary presence flag (zeroing historical x, y, heading, speed, and all a_{lon} inputs while retaining current speed), past-motion dropout

inspired by ChauffeurNet [1], removal of fallback clip-endpoint targets, and an initial 300-step gate-only warmup phase.

- **sig02.** Identical to `hist_ft`, but replacing the bisection-budgeted mask with an unconstrained sigmoid mask (initialized to a mean retention rate of 0.2) regularized by an explicit squared budget penalty $(\mathrm{mean}_p\, m_p - \rho)^2$ of weight 0.1.

Both fine-tuning variants (`hist_ft` and `sig02`) were trained for only 3 epochs.

Table 18: Closed-loop evaluation of unsubmitted variants (without AEB), paired as in Table 12 (comparing a single rollout of each variant against the two-rollout mean of the reference model).

Comparison	n	Ref.	Run	Diff.	95% CI	W/L	p
Baseline + TE4 vs. baseline (2)	150	0.479	0.504	+0.026	[-0.027, +0.079]	33/22	0.18
Mask + TE4 vs. mask (2)	150	0.517	0.495	-0.021	[-0.082, +0.041]	25/29	0.68
Mask + 8 samples vs. mask (2)	150	0.517	0.512	-0.005	[-0.051, +0.040]	24/21	0.77
hist_ft vs. mask (2)	150	0.517	0.495	-0.021	[-0.082, +0.037]	29/26	0.79
sig02, 1st vs. mask (2)	150	0.517	0.505	-0.012	[-0.077, +0.055]	27/28	1.00
sig02, 2nd vs. mask (2)	150	0.517	0.497	-0.020	[-0.084, +0.043]	29/29	1.00

I Runtime measurements

All local timing benchmarks were conducted on multi-tenant hosts shared with concurrent workloads and therefore differ from the official evaluation environment, which allocates 16 driver replicas across 4 GPUs with 2 concurrent rollouts per replica. Official runtime metrics are reported in the challenge submission status. Table 11 reports the end-to-end driver telemetry recorded during our local closed-loop evaluation runs (with video recording enabled and up to 12 concurrent rollouts per driver instance).

Table 19: Standalone runtime of the pre-configured submission container images launched without arguments in the challenge sandbox: 2 replicas with 1 active session each on a single shared GPU over an approximately 27 s rollout. All latencies are reported in seconds.

	Mean Drive	Planning call	Other calls, mean	Other calls, p50	Calls with detector
AEB 10 Hz	0.111	0.249	0.077	0.026	143/143
AEB 5 Hz	0.104	0.230	0.072	0.015	82/143

Table 20: Throughput saturation benchmark: logged rollout observations replayed continuously into 3 driver replicas (within the submission sandbox, video recording disabled) on a single shared NVIDIA RTX 6000 Ada GPU, with 2 concurrent sessions per replica and zero simulator rendering delay between consecutive RPC calls. Because eliminating simulator think-time exaggerates GPU contention, these values should be used only for relative comparisons across rows. All times are reported in seconds.

	Sessions	Ticks	Mean	p50	p90	p99	Per tick	Wall
No AEB	6	1071	0.294	0.001	1.457	1.512	0.322	64
AEB 10 Hz	6	1071	0.317	0.169	0.890	1.125	0.346	71
AEB 5 Hz	6	1071	0.306	0.113	1.167	1.365	0.333	67

In the back-to-back saturation benchmark (Table 20), enabling AEB increases the mean `Drive` call duration by 8% at 10 Hz and by 4% at 5 Hz relative to the policy without AEB. Note that the absolute latencies in this stress test do not reflect isolated model inference latency, as 6 sessions continuously contend for a single GPU with zero idle time between calls—unlike closed-loop simulation, where physics stepping and sensor rendering interleave with driver RPCs.

J Sources of the reported data

All open-loop metrics, training curves, and visualizations in Figs. 2–3 are generated directly from W&B runs `ccb412c9` and `03bdfc50` (project `starVLA_drive`), both initialized from commit `b5c4446` with random seed 42. All closed-loop evaluation metrics are extracted from each run's `aggregate/results-summary.json`, per-step `metrics.parquet`, and `rollout.asl` logs. The model code packaged in the deployed container images differs from the training commit solely by configuration flags that remain disabled for this checkpoint (specifically, ego-history dropout and fallback endpoint target filtering).